

Class 6: R Functions

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Q. Write a first function `generate_dna()` that returns a user specified length DNA sequence

```
generate_dna <- function(length) {  
  nuc <- c("A", "T", "G", "C")  
  sample(nuc, size = length, replace = TRUE)  
}
```

```
generate_dna(5)
```

```
[1] "A" "T" "G" "T" "T"
```

Q. Modify and improve our `generate_dna()` function to return it's generated sequence in a more standard format like "AGTAGTA" rather than te vector "A", "C", "G", "A".

```
generate_dna <- function(length, fasta) {  
  nuc <- c("A", "T", "G", "C")  
  if (fasta) {  
    sequence <- sample(nuc, size = length, replace = TRUE)  
    paste(sequence, collapse = "")  
  } else {  
    sample(nuc, size = length, replace = TRUE)  
  }  
}
```

```
generate_dna(5, TRUE)
```

```
[1] "TTGGC"
```

Q. Write a function called `generate_protein()`, that generates a user specified length protein sequence. Use that function to generate random protein sequences between length 6 and 12.

```
generate_protein <- function(length, space) {  
  aa <- c("A", "R", "N", "D", "C", "E", "Q", "G", "H", "I",  
         "L", "K", "M", "F", "P", "S", "T", "W", "Y", "V")  
  
  sequence <- sample(aa, size = length, replace = TRUE)  
  
  if (space) {  
    paste(sequence, collapse = "")  
  } else {  
    sequence  
  }  
}
```

Q. Use that function to generate random protein sequences between length 6 and 12.

```
for (i in 6:12) {  
  print(generate_protein(i, TRUE))  
}
```

```
[1] "FAHPRF"  
[1] "TPAKFNH"  
[1] "VICSPNNL"  
[1] "TKPNFCWFR"  
[1] "MTVWCEKKFV"  
[1] "HLKTLRWMEHL"  
[1] "CVNFWGWIYDSL"
```